Graph theory and extremal combinatorics

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L5-CO1 Level 5 Advanced · Combinatorics

Trees, bipartite graphs, the theorems of Mantel, Turán, Hall and Dirac, and small Ramsey numbers, with the extremal arguments that prove them and the olympiad problems they solve.
By the end you can translate a problem into a graph and use degrees, trees and odd cycles to prove structural facts; prove and apply Mantel, Turán and Hall, and find the matching extremal constructions; run longest-path, shortest-odd-cycle and maximum-degree arguments to olympiad standard.
Prerequisites: Invariants, monovariants and extremal arguments (L4-CO1), Recursion, bijections and double counting (L4-CO2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Among ten people there are 26 pairs of friends. Prove that some three of them are mutual friends.

Suppose not. Write $d(v)$ for the number of friends of v. If x and y are friends, they have no common friend, so $d(x)+d(y)\le 10$. Add this over all 26 friendships. Each person v appears in $d(v)$ of these inequalities, so the left-hand side adds up to $\sum_v d(v)^2$, and by the QM-AM inequality

$$\sum_v d(v)^2 \;\ge\; \frac{1}{10}\Big(\sum_v d(v)\Big)^2 = \frac{52^2}{10} > 260 = 26\cdot 10.$$

This is a contradiction. The same computation shows that a triangle-free graph on n vertices has at most $n^2/4$ edges (Mantel's theorem), and $26 > 100/4$.

The extremal principle (L4-CO1) and double counting (L4-CO2) are assumed.

1.1 Language

A graph G is a finite set V of vertices with a set E of edges, each edge an unordered pair of distinct vertices. Write $n=|V|$ and $e=|E|$. The degree $d(v)$ is the number of edges at v, and $N(v)$ is the set of neighbours of v. A walk is a sequence of vertices with consecutive ones adjacent, and a path is a walk with distinct vertices. A cycle is a path of at least three vertices whose ends are also adjacent, and its length is its number of edges. G is connected if any two vertices are joined by a path, and the maximal connected pieces are the components. An independent set contains no edge, and a clique contains every possible edge. K_n is the complete graph, and $K_{a,b}$ the complete bipartite graph with parts of sizes a and b.

Theorem 1 (Handshake lemma).
$\sum_{v\in V} d(v) = 2e$. In particular, the number of vertices of odd degree is even.

Proof. Each edge is counted once from each of its two ends. ■

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1.2 Trees

A tree is a connected graph with no cycle. A forest is a graph with no cycle, so its components are trees. A leaf is a vertex of degree 1.

Theorem 2.
(a) Every tree with $n \geq 2$ vertices has at least two leaves. (b) A tree with n vertices has $n-1$ edges, and a forest with n vertices and c components has $n-c$ edges. (c) A connected graph with n vertices and $n-1$ edges is a tree.

Proof. (a) Take a longest path $x_0x_1 \dots x_k$ (extremal principle). Then $k \geq 1$ since there is an edge. Every neighbour of x_0 lies on the path, or the path could be extended. If x_0 were adjacent to some x_i with $i \geq 2$, then $x_0x_1 \dots x_ix_0$ would be a cycle. So $d(x_0) = 1$, and similarly $d(x_k) = 1$.

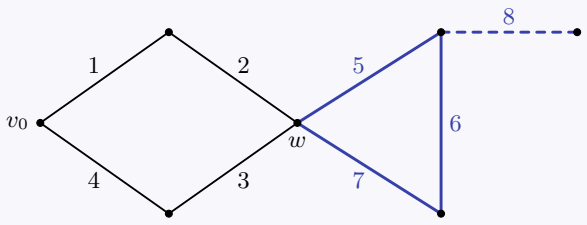
(b) Induct on n. Deleting a leaf from a tree with $n \geq 2$ vertices leaves a tree with one vertex and one edge fewer. For a forest, add the formula over the components.

(c) While the graph has a cycle, delete one edge of a cycle. This keeps the graph connected, since the rest of the cycle still joins the two ends. We finish with a tree, which by (b) has $n-1$ edges, so no edge was deleted. ■

The proof of (c) shows that every connected graph contains a spanning tree, that is, a tree on all n vertices. The longest-path argument of (a) proves more. If every vertex has degree at least $d \geq 2$, then the end x_0 has at least d neighbours on the path, the furthest being some x_i with $i \geq d$, so G has a cycle of length at least $d+1$.

Example 1.
(IMO 1991, Problem 4) Let G be a connected graph with k edges. Prove that the edges can be labelled $1, 2, \dots, k$ so that at every vertex of degree at least 2 the greatest common divisor of the labels of the edges at that vertex is 1.

Solution.
Consecutive integers are coprime, and 1 is coprime to everything. So we only need each vertex of degree at least 2 to meet either the edge labelled 1 or two edges with consecutive labels.
Start at any vertex v_0 and walk along edges, never using an edge twice, labelling the edges $1, 2, 3, \dots$ in the order they are used, until the walk stops at a vertex with no unused edge. Suppose unused edges remain. A path from v_0 to an end of one of them shows that some unused edge has an end w already visited, since every used edge has both ends visited. Start a new walk at w along unused edges, continuing the labels, and repeat until every edge is labelled.



Let $v \neq v_0$ have degree at least 2. Every new walk starts at a vertex already visited, so v is visited for the first time when a walk arrives at v along an edge. At that moment only one

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edge at v has been used, so the walk does not stop and leaves along a second edge. The two edges have consecutive labels. Finally v_0 meets the edge labelled 1. ■

Remark. The condition asks for very little, since two consecutive labels at a vertex are enough. Labels in the order of a walk give consecutive labels wherever the walk passes through a vertex, so the only places to check are those where a walk starts or stops.

1.3 Bipartite graphs and odd cycles

G is bipartite if V splits into two independent sets, that is, if G has a proper colouring with two colours.

Theorem 3.
A graph is bipartite if and only if the graph has no cycle of odd length.

Proof. A cycle in a bipartite graph alternates between the parts, so the cycle has even length. Conversely, suppose there is no odd cycle. First, a closed walk of odd length contains an odd cycle. If no vertex repeats, the walk is a cycle. If some vertex repeats, the walk splits at that vertex into two shorter closed walks whose lengths add to an odd number, so one of them is odd, and we induct on the length. Now work in one component, fix a vertex r, and colour each vertex by the parity of its distance from r. If two adjacent vertices u, v had the same colour, then a shortest path from r to u, the edge uv and a shortest path from v back to r would form a closed walk of odd length $\operatorname{dist}(r,u) + \operatorname{dist}(r,v) + 1$, hence an odd cycle. ■

In extremal problems a *shortest* odd cycle is often the right object to take, since such a cycle has no chords. A chord would split the cycle into two shorter cycles, one of them odd.

1.4 Mantel and Turán

Theorem 4 (Mantel).
A triangle-free graph on n vertices has at most $\lfloor n^2/4 \rfloor$ edges, with equality only for $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$.

Proof. Let v have the largest degree Δ. Since G has no triangle, $N(v)$ is independent, so every edge has an end in $S = V \setminus N(v)$, a set of $n - \Delta$ vertices each of degree at most Δ. Hence

$$e \;\le\; \sum_{s \in S} d(s) \;\le\; (n-\Delta)\Delta \;\le\; \frac{n^2}{4}.$$

For equality every inequality must be tight. No edge has both ends in S, every $s \in S$ has degree $\Delta = |N(v)|$ and so is adjacent to all of $N(v)$, and $\Delta(n-\Delta) = \lfloor n^2/4 \rfloor$ forces $\{\Delta, n-\Delta\} = \{\lfloor n/2 \rfloor, \lceil n/2 \rceil\}$. So G is complete bipartite with parts S and $N(v)$. ■

Theorem 5 (Turán).
If a graph on n vertices contains no K_{r+1}, then $e \le \left(1 - \frac{1}{r}\right) \frac{n^2}{2}$.

Proof. Induct on r. For $r = 1$ there are no edges. Let $r \ge 2$ and let v have the largest degree Δ. The graph induced on $N(v)$ has no K_r (with v this would give a K_{r+1}), so by induction this graph has

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at most $\left(1-\frac{1}{r-1}\right)\frac{\Delta^2}{2}$ edges. Every other edge has an end in $V \setminus N(v)$, so there are at most $(n-\Delta)\Delta$ of them. Therefore

$$e \;\le\; \frac{r-2}{r-1}\cdot\frac{\Delta^2}{2} + (n-\Delta)\Delta \;=\; n\Delta - \frac{r}{2(r-1)}\Delta^2 \;\le\; \frac{r-1}{r}\cdot\frac{n^2}{2},$$

where the last step is the maximum of a quadratic in Δ, attained at $\Delta = \frac{r-1}{r}n$. ■

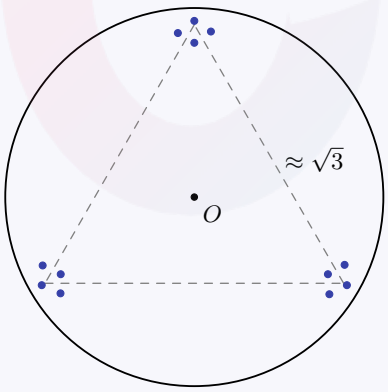
The bound is attained when $r \mid n$ by the complete r-partite graph with equal parts. For general n the maximum is attained by the complete r-partite graph with parts as equal as possible.

Example 2.
Twelve distinct points lie in a closed disc of radius 1. At most how many of the 66 pairs of points can be at distance greater than $\sqrt{2}$?

Solution.
Join two points when their distance exceeds $\sqrt{2}$. We show that no four points are pairwise joined. If one of four points is the centre O, then O is within 1 of the others. Otherwise the four rays from O through the points divide the full angle, so two of them make an angle $\theta \le 90^\circ$. For those two points P, Q, the cosine rule gives

$$PQ^2 = OP^2 + OQ^2 - 2\,OP\cdot OQ\cos\theta \;\le\; OP^2 + OQ^2 \;\le\; 2.$$

So the graph has no K_4, and Theorem 5 with $r = 3$ gives at most $\frac{2}{3}\cdot\frac{144}{2} = 48$ edges.
The equality case of Turán's theorem says what to build, namely three groups of four with every pair from different groups far apart. Put four points very close to each vertex of an equilateral triangle inscribed in the circle. Points in different groups are at distance close to $\sqrt{3} > \sqrt{2}$, so $3\cdot 4\cdot 4 = 48$ pairs are far apart. The answer is 48.



Remark. The number $\sqrt{2}$ is the side of the inscribed square, which is why we tested four points. When a problem asks for the largest number of pairs with some property, find the largest set in which every pair has the property. If that set has r elements, Theorem 5 gives the bound and the complete r-partite graph is the example to aim for.

Counting pairs of neighbours gives a bound of a different kind.

Theorem 6 (no 4-cycle).
If a graph on n vertices has no cycle of length 4, then $e \le \frac{n}{4}\left(1+\sqrt{4n-3}\right)$.

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Proof. Count the pairs $\{x, y\}$ of neighbours of a vertex v, over all v. Two vertices x, y with two common neighbours would lie on a 4-cycle, so each pair is counted at most once and $\sum_v \binom{d(v)}{2} \leq \binom{n}{2}$. By Theorem 1 and the QM-AM inequality, $\sum_v d(v)^2 \geq \frac{4e^2}{n}$. Hence $\frac{4e^2}{n} - 2e \leq n(n-1)$, and solving the quadratic inequality for e gives the bound. ■

1.5 Hall's theorem

A matching is a set of edges with no common ends. In a bipartite graph with parts A and B, for $S \subseteq A$ write $N(S)$ for the set of vertices of B adjacent to some vertex of S.

Theorem 7 (Hall).
A bipartite graph with parts A, B has a matching covering every vertex of A if and only if $|N(S)| \geq |S|$ for every $S \subseteq A$.

Proof. The condition is necessary, since the partners of S are $|S|$ distinct vertices of $N(S)$. For sufficiency, induct on $|A|$. The case $|A| = 1$ is clear.

Case 1. Here $|N(S)| \geq |S| + 1$ for every non-empty $S \subsetneq A$. Match any $a \in A$ with any neighbour b and delete both. In the remaining graph each non-empty $S \subseteq A \setminus \{a\}$ has lost at most the one neighbour b, so S still has at least $|S|$ neighbours, and induction finishes.

Case 2. Some non-empty $S_0 \subsetneq A$ has $|N(S_0)| = |S_0|$. Hall's condition holds inside the graph on $S_0 \cup N(S_0)$, so by induction S_0 can be matched into $N(S_0)$. In the graph on $(A \setminus S_0) \cup (B \setminus N(S_0))$, a set $T \subseteq A \setminus S_0$ has neighbourhood $N(T \cup S_0) \setminus N(S_0)$, of size at least $|T| + |S_0| - |S_0| = |T|$. So induction matches $A \setminus S_0$ there too, and the two matchings together cover A. ■

Corollary 8.
If every vertex of a bipartite graph (multiple edges allowed) has degree $k \geq 1$, then the graph has a perfect matching.

Proof. The $k|S|$ edges leaving $S \subseteq A$ all end in $N(S)$, which receives $k|N(S)|$ edges in total, so $|N(S)| \geq |S|$. Also $k|A| = e = k|B|$, so a matching covering A is perfect. ■

Example 3.
(Putnam 2012, B3) A tournament of $2n$ teams lasted $2n - 1$ days. On each day every team played one game against another team, with one team winning and one losing in each of the n games, and over the tournament each team played every other team exactly once. Prove that one can choose a winning team from each day without choosing any team twice.

Solution.
Choosing greedily day by day fails, since an early choice can use up the only winner that a later day could offer. This is a matching problem. Let A be the set of days and B the set of teams, and join a day to each team that won on that day. We need a matching covering A. Suppose Hall's condition fails for a set S of days, so that $|N(S)| < |S| \leq 2n - 1$. Then some team t is not in $N(S)$, which means that t lost on every day of S. The teams that beat t on those days are $|S|$ different teams, because t meets each team only once, and each of them won on a day of S. So $|N(S)| \geq |S|$, a contradiction. By Theorem 7 the matching exists. ■
Remark. To verify Hall's condition you need a lower bound for $|N(S)|$. The usual sources are a count of edges, as in Corollary 8, or a single vertex outside $N(S)$, as here. A team outside $N(S)$ lost on every day of S, and its opponents on those days are forced into $N(S)$.

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1.6 Ramsey numbers

$R(s,t)$ is the least n such that every red/blue colouring of the edges of K_n has a red K_s or a blue K_t.

Theorem 9.
$R(3,3) = 6$, and $R(s,t) \le R(s-1,t) + R(s,t-1)$ for $s,t \ge 2$. Hence $R(s,t) \le \binom{s+t-2}{s-1}$.

Proof. Let $n = R(s-1,t) + R(s,t-1)$ and fix a vertex v of a coloured K_n. Its $n-1$ edges split into r red and b blue with $r + b = n - 1$, so $r \ge R(s-1,t)$ or $b \ge R(s,t-1)$. In the first case the red neighbours of v contain a blue K_t, or a red K_{s-1} which with v is a red K_s. The second case is symmetric. The binomial bound follows by induction on $s+t$ from $R(s,1) = R(1,t) = 1$ and Pascal's rule. With $R(2,3) = R(3,2) = 3$ the recursion gives $R(3,3) \le 6$, and the pentagon with red sides and blue diagonals shows $R(3,3) > 5$. ■

The values worth remembering are $R(3,3) = 6$, $R(3,4) = 9$, $R(3,5) = 14$ and $R(4,4) = 18$. Theorem 9 gives only $R(3,4) \le 10$, and parity improves this to 9. In a colouring of K_9 with no red triangle and no blue K_4, the red neighbours of a vertex form a blue clique, so there are at most 3 of them. Its blue neighbours contain no triangle of either colour, so there are at most 5. Hence every vertex has exactly 3 red edges, and there are $\frac{9\cdot 3}{2}$ red edges, which is absurd. Practice problem 2 uses the same method. That $R(4,4) > 17$ needs a specific colouring of K_{17} (Greenwood and Gleason, 1955), which I will not reproduce. Lower bounds for large k come from the counting argument in L5-CO2.

1.7 Extremal arguments on graphs

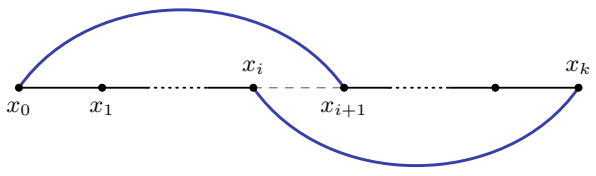
So far the extreme objects were a longest path, a vertex of largest degree and a shortest odd cycle. A longest path can also be rearranged.

Theorem 10 (Dirac).
If $n \ge 3$ and every vertex has degree at least $\frac{n}{2}$, then G has a cycle through all n vertices.

Proof. Two non-adjacent vertices have at least n neighbours in total among the other $n-2$ vertices, so they have a common neighbour, and G is connected. Take a longest path $x_0 x_1 \dots x_k$. All neighbours of x_0 and of x_k lie on the path. Let I be the set of i with x_0 adjacent to x_{i+1}, and J the set of i with x_k adjacent to x_i. Both are subsets of $\{0,1,\dots,k-1\}$, and $|I| + |J| \ge n > k$, so some i lies in both. Then

$$x_0 x_1 \dots x_i \, x_k x_{k-1} \dots x_{i+1} \, x_0$$

is a cycle C through all $k+1$ vertices of the path.



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If $k+1 < n$, then by connectivity some vertex y outside C is adjacent to a vertex of C. Opening C at that vertex and adding y gives a path with $k+2$ vertices, a contradiction. So C passes through every vertex. ■

The bound is sharp, since $K_{m,m+1}$ has every degree at least $\frac{n-1}{2}$ and no such cycle.

Another useful extreme object is the subgraph left after repeatedly deleting vertices of small degree.

Example 4.
A graph has 30 vertices and 121 edges. Prove that the graph contains a copy of every tree with 5 edges.

Solution.
Since $121 > 4 \cdot 30$, we guess the general statement that *a graph with n vertices and more than $(k-1)n$ edges contains every tree with k edges.*
Step 1. Repeatedly delete a vertex whose degree in the current graph is at most $k-1$. If every vertex were deleted, all edges would be removed, at most $k-1$ with each deletion, so $e \le (k-1)n$, a contradiction. So the process stops at a non-empty subgraph H in which every degree is at least k.
Step 2. Let T be a tree with k edges, hence $k+1$ vertices. Order its vertices $t_0, t_1, \dots, t_k$ so that each t_i with $i \ge 1$ has exactly one neighbour among $t_0, \dots, t_{i-1}$ (repeatedly remove a leaf, and reverse the order of removal). Place t_0 at any vertex of H. When $t_0, \dots, t_{i-1}$ have been placed at distinct vertices, t_i must go to a neighbour of the image of its earlier neighbour t_j. That image has at least k neighbours in H, and at most $i-1 \le k-1$ of them are already used (all placed vertices except the image of t_j itself). So a free neighbour exists, and the embedding never gets stuck. ■
Remark. A bound on the number of edges controls only the average degree, and a greedy embedding needs every degree to be large. Step 1 turns the first into the second, and is worth trying whenever a problem gives only the number of edges. The constant $k-1$ is not best possible. A conjecture of Erdős and Sós asserts that more than $\frac{k-1}{2}n$ edges suffice.

Common traps.

- Assuming a graph is connected. Theorem 2(b) has $n-c$, not $n-1$, for forests.
- Proving the bound in an extremal problem and forgetting the construction, or the reverse. "Find the maximum" needs both.
- Applying Hall to the wrong side. The condition is checked for subsets of the side you want covered.
- In a longest path argument, forgetting to say why every neighbour of the end lies on the path.
- Quoting Turán with the wrong r. No K_{r+1} gives $(1-\frac{1}{r})\frac{n^2}{2}$.

1.8 How to spot this in a contest

Whenever a problem has people and friendships, cities and roads, or objects and a symmetric relation, draw the graph and translate. The standard moves, in the order I try them, are:

1. count degrees (handshake lemma, double counting edges or pairs of neighbours),

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2. take an extreme object, for example a longest path, a shortest odd cycle, a vertex of largest or smallest degree, or the subgraph left after deleting vertices of small degree,
3. if the question is about two-colouring or splitting into two groups, look for odd cycles (Theorem 3),
4. if a forbidden subgraph is a triangle or a clique, compare with Mantel or Turán, and expect the extremal example to be complete multipartite,
5. if each object must be assigned its own distinct partner, set up a bipartite graph and check Hall's condition,
6. if a statement is "every colouring contains a monochromatic ...", fix a vertex and use pigeonhole on its edges, as in Theorem 9.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆⋆ Eight teams play a tournament lasting seven days. On each day every team plays one game, and over the seven days every two teams meet exactly once. A set of four teams is called *spread* if the six games among those four teams are played on six different days. Find the smallest possible number of spread sets.

Problem 2 ⋆⋆⋆ Prove that among any seven people there are three mutual friends or four people among whom at most one pair are friends. Show that this is false for six people.

Problem 3 ⋆⋆⋆ Twelve distinct lines are given in space. Two lines are called perpendicular if their directions are at right angles, whether or not the lines meet. Find the largest possible number of sets of three of the lines that are pairwise perpendicular.

Problem 4 ⋆⋆⋆ A country has $2m$ towns and $2m-1$ roads. Each road joins two towns, and one can travel between any two towns along the roads. The *route* between two towns is the shortest path of roads joining them, and its length is its number of roads. The towns are to be paired off into m pairs of twin towns. Prove that this can be done so that some town lies on all m routes between twins, possibly as an end of a route. Prove also that every pairing with this property has the largest total length of the m routes among all pairings.

Problem 5 ⋆⋆⋆⋆ In a competition every contestant solved at least one problem. Whenever a contestant solved a problem, the number of problems solved by that contestant was at least the number of contestants who solved that problem. Prove that each contestant can be awarded a prize for one of the problems that the contestant solved, with no two prizes awarded for the same problem. Prove also that if the number of problems equals the number of contestants, then whenever a contestant solved a problem, the number of problems solved by that contestant was equal to the number of contestants who solved that problem.

Problem 6 ⋆⋆⋆⋆ At King Arthur's court there are $2n$ knights, where $n \geq 2$. Any two knights are either friends or enemies, and every knight has at most $n-2$ enemies. Sir Lancelot wishes to sit between two given friends of his, Sir Gawain and Sir Percival. Prove that the knights can be seated at a round table so that this wish is granted and no two enemies sit next to each other.

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Show also that if every knight has at most $n-1$ enemies, such a seating need not exist, even if Lancelot asks only to sit next to Gawain.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **IMO 1964 Problem 4** ⋆⋆
   Compare with the proof that $R(3,3)=6$.
2. **BMO2 2009/10 Problem 1** ⋆⋆⋆
   If two children are at distance k in the friendship graph, how far apart can they stand in the line?
3. **USAMO 2001 Problem 1** ⋆⋆⋆
   Count pairs of balls sharing a box in two ways, then construct an example meeting the bound.
4. **BMO2 2005/06 Problem 4** ⋆⋆⋆
   For each question, look at the set of children who got that question wrong, and see what the condition says about two such sets.
5. **APMO 1990 Problem 4** ⋆⋆⋆
   Fix a vertex and compare the degrees of two adjacent vertices by matching up their other neighbours. Then count the vertices at distance 1 and 2.
6. **USAJMO 2022 Problem 2** ⋆⋆⋆
   Think of a choice of cells, no two in a row or column, as a matching between rows and columns.
7. **BMO2 2013/14 Problem 1** ⋆⋆⋆
   For the lower bound, find a large family of diagonals that pairwise cross. Then look for a colouring that matches this family.
8. **IMO 1992 Problem 3** ⋆⋆⋆⋆
   Use $R(3,3)=6$, and think about how few edges can be left uncoloured.
9. **BMO2 2004/05 Problem 4** ⋆⋆⋆⋆
   Consider how many sets the most popular element lies in, and split into cases.
10. **USAMO 2007 Problem 4** ⋆⋆⋆⋆
    Think of the animal as a connected graph on its cells and look for a good place to cut the graph.
11. **IMO 2007 Problem 3** ⋆⋆⋆⋆
    Put a largest clique in one room and move its members across one at a time. Stop when the two rooms are level or one apart, and in the second case look for competitors to send back.
12. **IMO 2019 Problem 3** ⋆⋆⋆⋆
    Find properties of the graph that the moves can be made to preserve, and reduce the number of vertices one at a time.

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4 Hints

1. If a set of four teams is not spread, two of its six games are played on the same day. Count the pairs of games played on the same day, and ask how many sets of four teams one such pair can spoil.

2. Assume not. Bound the number of friends and the number of strangers of one person, as in the proof that $R(3,4) \leq 9$, and then look for a person with few friends.

3. Make a graph on the lines. Why does this graph contain no K_4? To bound the number of triangles, delete the three vertices of one triangle and count the triangles that are lost, using Theorem 5.

4. Removing a town c breaks the network into pieces. Choose c so that the largest piece is as small as possible, and pair towns from different pieces. For the second part, bound the number of routes that can use a given road.

5. Join each contestant to the problems that the contestant solved. To compare a set S of contestants with the set of problems they solved, give the edge from a contestant c to a problem the weight $\frac{1}{d(c)}$ and add up the weights of the edges at S.

6. Follow the proof of Theorem 10, but only among paths that contain the two edges from Lancelot to Gawain and to Percival. Both the cycle and the longer path that the proof builds must keep these two edges.

5 Solutions

Problem 1

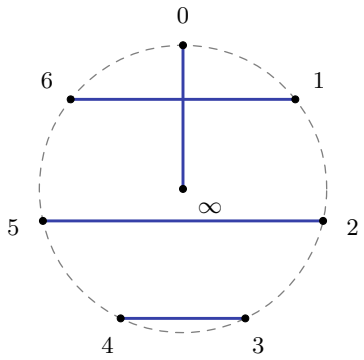
Eight teams play a tournament lasting seven days. On each day every team plays one game, and over the seven days every two teams meet exactly once. A set of four teams is called spread *if the six games among those four teams are played on six different days. Find the smallest possible number of spread sets.*

The answer is 28. Two games on the same day have no team in common, since each team plays once a day. So two games on the same day involve four teams, and we call this pair of games a *clash* in that set of four teams. A set of four teams is spread exactly when the set contains no clash.

Each day has 4 games and hence $\binom{4}{2} = 6$ clashes, so there are $7 \cdot 6 = 42$ clashes in all. Each clash lies in exactly one set of four teams. So at most 42 of the $\binom{8}{4} = 70$ sets are not spread, and at least 28 sets are spread.

For equality we need a schedule in which no set of four teams contains two clashes. Label the teams $\infty, 0, 1, \ldots, 6$, with the numbers taken modulo 7. On day i, for $i = 0, 1, \ldots, 6$, let ∞ play i, and let $i + t$ play $i - t$ for $t = 1, 2, 3$.

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The teams i, $i+t$ and $i-t$ with $t=1,2,3$ are all seven numbered teams, so every team plays once on day i. Two numbered teams $a \neq b$ meet on day i exactly when $a+b \equiv 2i \pmod 7$, which holds for exactly one i, and ∞ meets i on day i only. So every two teams meet exactly once.

Suppose a set of four teams contains two clashes. They are two different ways to split the four teams into two games, so we can name the teams a, b, c, d so that a plays b and c plays d on a day i, while a plays c and b plays d on a day j. Here $i \neq j$, because a plays once a day. Each of the four teams is in one game of each clash, so if ∞ is among them we may take $a=\infty$.

If all four teams are numbered, then $a+b \equiv c+d \equiv 2i$ and $a+c \equiv b+d \equiv 2j$. Subtracting gives $b-c \equiv 2i-2j$ and $c-b \equiv 2i-2j$, so $2(b-c) \equiv 0$ and $b=c$, a contradiction. If $a=\infty$, then $b=i$ and $c=j$, and the other two games give $j+d \equiv 2i$ and $i+d \equiv 2j$. Subtracting gives $3(i-j) \equiv 0$, so $i=j$, a contradiction.

Hence in this schedule the 42 clashes lie in 42 different sets, and exactly $70-42=28$ sets are spread.

Answer: 28

Problem 2

Prove that among any seven people there are three mutual friends or four people among whom at most one pair are friends. Show that this is false for six people.

Call a set of four people *sparse* if at most one pair among them are friends. Suppose seven people have no three mutual friends and no sparse set, and fix a person v.

The friends of v are pairwise strangers, since two friends of v who were friends would form three mutual friends with v. Four friends of v would form a sparse set, so v has at most 3 friends.

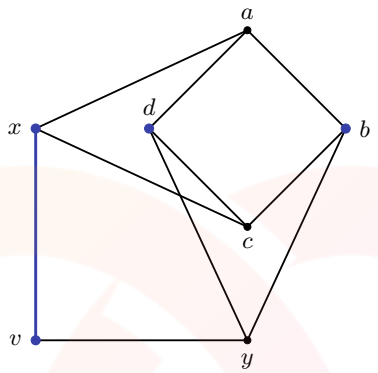
Let S be the set of strangers of v. If some $x \in S$ were a stranger to two other members y, z of S, then the five pairs vx, vy, vz, xy, xz would be strangers and $\{v,x,y,z\}$ would be sparse. So the pairs of strangers inside S are disjoint. If $|S| \geq 5$, take five members of S. At most two disjoint pairs of strangers lie among them, so taking one person from each such pair together with the people in no such pair, we find three of the five who are mutual friends. Hence $|S| \leq 4$, and v has at least $6-4=2$ friends. Moreover, if $|S|=4$ then S contains exactly two disjoint pairs of strangers (with at most one, three members of S would be mutual friends), so the friendships inside S form a 4-cycle $abcd$, with a, c strangers and b, d strangers.

So every person has 2 or 3 friends. They cannot all have 3, since the number of pairs of friends would then be $\frac{7\cdot 3}{2}$. Now fix a person v with exactly 2 friends x and y. They are strangers, and the 4 strangers of v form a 4-cycle of friendships $abcd$ as above.

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Each of a, b, c, d has two friends on the cycle and at most 3 friends in all, so at most one friend in $\{x, y\}$. Suppose a has no friend in $\{x, y\}$. Then the strangers of a are v, c, x, y, so by the third paragraph, applied to a in place of v, their friendships form a 4-cycle. Since v, c are strangers and x, y are strangers, this cycle is $vxcy$, and c is friends with b, d, x and y, which is too many. Hence each of a, b, c, d has exactly one friend in $\{x, y\}$, which gives four friendships between $\{a, b, c, d\}$ and $\{x, y\}$.

Now x is a friend of v and has at most 3 friends, so x has at most 2 friends among a, b, c, d, and the same holds for y. Therefore each of x, y has exactly 2 friends among a, b, c, d. Two friends of x are not friends of each other, so they are opposite on the cycle. Say x is friends with a and c. Then in $\{v, x, b, d\}$ the only pair of friends is v, x, so this set is sparse, a contradiction.



For six people, split them into two groups of three and let two people be friends exactly when they are in different groups. Any three people include two from the same group, so no three are mutual friends, and four people split between the groups as $2 + 2$ or $3 + 1$, so they include 4 or 3 pairs of friends.

Problem 3

Twelve distinct lines are given in space. Two lines are called perpendicular if their directions are at right angles, whether or not the lines meet. Find the largest possible number of sets of three of the lines that are pairwise perpendicular.

The answer is 64. Make a graph G whose vertices are the lines, joining two lines when they are perpendicular. We want the largest possible number of triangles.

G has no K_4. Suppose four of the lines are pairwise perpendicular, with direction vectors u_1, u_2, u_3, u_4. If $au_1 + bu_2 + cu_3 = 0$, then the scalar product with u_1 gives $a|u_1|^2 = 0$, so $a = 0$, and similarly $b = c = 0$. So u_1, u_2, u_3 span the space and $u_4 = au_1 + bu_2 + cu_3$ for some a, b, c. Then $|u_4|^2 = u_4 \cdot (au_1 + bu_2 + cu_3) = 0$, which is absurd.

Claim. A graph with N vertices and no K_4 has at most $\frac{N^3}{27}$ triangles. We use induction on N. A graph with no triangle satisfies the claim, and this covers $N \leq 2$. Otherwise let xyz be a triangle, and let G' be the graph left when x, y, z are deleted. Sort the triangles of G by the number of their vertices among x, y, z.

No vertex. These are the triangles of G', and there are at most $\frac{(N-3)^3}{27}$ of them by induction.

One vertex. Such a triangle consists of an edge uv of G' and one of x, y, z adjacent to both u and v. Two of x, y, z adjacent to both u and v would form a K_4 with u and v. So each edge of G' gives at most one triangle, and by Theorem 5 with $r = 3$ there are at most $\frac{(N-3)^2}{3}$ of them.

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Two vertices. The third vertex u is adjacent to two of x, y, z and cannot be adjacent to all three, so each of the $N-3$ vertices of G' gives at most one triangle.

Three vertices. Only xyz.

In total the number of triangles is at most

$$\frac{(N-3)^3}{27}+\frac{(N-3)^2}{3}+(N-3)+1=\left(\frac{N-3}{3}+1\right)^3=\frac{N^3}{27}.$$

With $N=12$ there are at most 64 sets.

For an example with 64, take three pairwise perpendicular directions and four lines parallel to each. Two of the lines are perpendicular exactly when they are in different groups, so the sets of three pairwise perpendicular lines are the choices of one line from each group, and there are $4\cdot 4\cdot 4=64$ of them.

Answer: 64

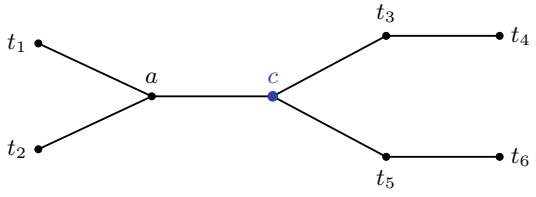
Problem 4

A country has $2m$ *towns and* $2m-1$ *roads. Each road joins two towns, and one can travel between any two towns along the roads. The* route *between two towns is the shortest path of roads joining them, and its length is its number of roads. The towns are to be paired off into* m *pairs of twin towns. Prove that this can be done so that some town lies on all* m *routes between twins, possibly as an end of a route. Prove also that every pairing with this property has the largest total length of the* m *routes among all pairings.*

By Theorem 2(c) the network is a tree, so any two towns are joined by exactly one path, which is the route. Removing a town c leaves several trees, which we call the *branches* at c. Two towns in different branches at c have a route through c.

Step 1. Some town c has no branch with more than m towns. Choose c so that the largest branch at c has as few towns as possible, and suppose that this branch B has more than m towns. Let c' be the neighbour of c in B. One branch at c' consists of the towns outside B, and there are fewer than m of these. Every other branch at c' lies inside B and does not contain c', so has fewer towns than B. Thus the largest branch at c' is smaller than B, which contradicts the choice of c.

Step 2. The pairing. Take c as in Step 1 and pair c with a town a of a largest branch. The other $2m-2$ towns now lie in sets of at most $m-1$ towns each, one set for each branch. Indeed the branch of a has lost a town, and two branches with m towns each cannot exist among $2m-1$ towns. List these towns as $t_1, t_2, \ldots, t_{2m-2}$, one branch after another, and pair t_i with t_{i+m-1} for $i=1,\ldots,m-1$. The towns of one branch occupy at most $m-1$ consecutive places of the list, so t_i and t_{i+m-1} are in different branches and their route passes through c. The route from c to a ends at c.



Step 3. Removing a road e splits the towns into two sets. Let $s(e)$ be the number of towns in the smaller set. In any pairing, a route that uses e joins a town of the smaller set to its twin, so at most $s(e)$ routes use e. Adding over the roads, the total length of the routes is at most $\sum_e s(e)$.

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Now take a pairing in which a town c lies on every route, and a road e. Let U be the set of towns on the side of e that does not contain c. For each town u of U the route from u to its twin contains c, so this route uses e. A route between two towns of U stays inside U and misses c, so no two towns of U are twins, and the $|U|$ routes are different. Hence $|U| \le s(e)$, so U is a smaller side, and exactly $s(e)$ routes use e. The total length is therefore $\sum_e s(e)$, the largest possible. ■

Problem 5

In a competition every contestant solved at least one problem. Whenever a contestant solved a problem, the number of problems solved by that contestant was at least the number of contestants who solved that problem. Prove that each contestant can be awarded a prize for one of the problems that the contestant solved, with no two prizes awarded for the same problem. Prove also that if the number of problems equals the number of contestants, then whenever a contestant solved a problem, the number of problems solved by that contestant was equal to the number of contestants who solved that problem.

Make a bipartite graph, joining a contestant c to a problem p when c solved p. Then $d(c) \ge 1$ for every contestant, and $d(c) \ge d(p)$ for every edge cp. The prizes are a matching that covers every contestant, so by Theorem 7 we must show that $|N(S)| \ge |S|$ for every set S of contestants.

Give each edge cp the weight $\frac{1}{d(c)}$. The $d(c)$ edges at a contestant c have total weight 1, so the edges at S have total weight $|S|$. Every such edge cp ends at a problem p of $N(S)$ and has weight $\frac{1}{d(c)} \le \frac{1}{d(p)}$. A problem p lies on $d(p)$ edges, so the edges from S to p have total weight at most $d(p) \cdot \frac{1}{d(p)} = 1$. Adding over the problems of $N(S)$ gives $|S| \le |N(S)|$.

For the second part, let there be k contestants and k problems, and let W be the total weight of all the edges. As above $W = k$. On the other hand, the edges at any one problem have total weight at most 1, so $W \le k$, with equality only if every edge cp has weight exactly $\frac{1}{d(p)}$. Since $W = k$, equality holds, and $d(c) = d(p)$ for every edge cp. ■

Problem 6

At King Arthur's court there are $2n$ knights, where $n \ge 2$. Any two knights are either friends or enemies, and every knight has at most $n-2$ enemies. Sir Lancelot wishes to sit between two given friends of his, Sir Gawain and Sir Percival. Prove that the knights can be seated at a round table so that this wish is granted and no two enemies sit next to each other. Show also that if every knight has at most $n-1$ enemies, such a seating need not exist, even if Lancelot asks only to sit next to Gawain.

Let G be the graph of friendships on the $2n$ knights. Every vertex has degree at least $2n-1-(n-2) = n+1$. Write A for Lancelot and B, C for Gawain and Percival, and call AB and AC the *fixed* edges. A seating as required is a cycle through all $2n$ vertices that contains both fixed edges.

Among all paths that contain both fixed edges (the path BAC is one) take a longest, $x_0x_1 \dots x_k$. A neighbour of x_0 outside the path could be placed before x_0, so every neighbour of x_0 lies on the path, and likewise for x_k. In particular $k \ge n+1$.

Let I be the set of i with x_0 adjacent to x_{i+1}, and J the set of i with x_k adjacent to x_i. Both are subsets of $\{0, 1, \dots, k-1\}$, and $|I| + |J| \ge 2n+2 \ge k+3$, so at least three indices lie in both. For at most two indices i is x_ix_{i+1} a fixed edge. Choose i in both sets with x_ix_{i+1} not fixed. Then

$$x_0x_1 \dots x_i\, x_k x_{k-1} \dots x_{i+1}\, x_0$$

is a cycle Z through the vertices of the path, and Z contains every edge of the path except possibly x_ix_{i+1}. So Z contains both fixed edges.

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Suppose some vertex is not on Z. Since Z has $k+1 \geq n+2$ vertices, at most $n-2$ vertices are not on Z. Let y be one of them. At most $n-3$ neighbours of y are off Z, so y has at least four neighbours on Z. Let $z \neq A$ be one of them. Only A lies on both fixed edges, so one of the two edges of Z at z is not fixed. Remove that edge from Z and add the edge zy. The result is a path with $k+2$ vertices that contains both fixed edges, a contradiction. So Z passes through every vertex.

For the second part, split the knights into two groups of n, with Lancelot and Gawain in the same group. Let two knights be friends exactly when they are in different groups, except that Lancelot and Gawain are also friends. Every knight has at most $n-1$ enemies. Suppose there is a seating with no two enemies adjacent. Each knight of the other group sits between two knights of the group of Lancelot. This uses $2n$ places next to knights of that group, which is all of them. So no two knights of that group are neighbours, and Lancelot does not sit next to Gawain.

The online version, with progress tracking, is at kevinvariant08.github.io/pathways.html.

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